Date: 2025-05-07 Project No./ID: D1-0038

Technical Design Specification

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Primary location: Alimak Group

Project Title
New Door/Gate Lock

Type of product/component
Locking mechanism

Applicable codes/regulations
En-12159, EN-8120, EN-8150, Machine directive, Hiss-direktivet, ISO 14119, ISO 13849-1, IEC 62061, IEC 61508, IEC 61000-4-x, EN 60204-1

Non-Compliance on regulations

See Appendix B, “Norm requirements”

Environmental conditions
High humidity, sub-zero temperatures, and heated conditions (-20°C to +40°C), IP65

Sizes and dimensions
245x82x44, See appendix for DRW´s

Weight
2 kg

Compatible systems (As of today)
Scando 650A, and newer system updates.

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1. Summary

The Door Lock Module is a safety-critical, electromechanical device designed for integration in rack and pinion lifts and hoists. Its primary purpose is to manage secure access to critical access points such as doors, roof hatches, and inspection covers, ensuring personnel safety and machine integrity. The lock will perform in accordance with PL-d safety requirements.
The purpose of the product is to update the existing locks and mechanisms that operate on the doors and landings today. Also to align the Construction and Industrial hoist components as far as possible.
The project will deliver a complete lock and pin that fulfills the requirements of the new and existing industrial and construction hoist norms.

2. Background and Need

The existing locks on our hoists/doors will not fulfill the upcoming norm requirements and needs to be updated to meet this new requirement. The project also predicts a cost reduction with this in comparison to purchasing a complete lock system from a supplier.

3. Functional description

The lock consists of a bi-stable solenoid-driven shaft mechanism, which either engages (locks) or disengages (unlocks) a lock pin inserted into the main body.
- The locking shaft position is secured mechanically using permanent magnets at each end position (locked/unlocked) and engages 6 mm into the lock pin.

- Locking/unlocking is triggered via an external I/O signal.
- Redundant Hall sensors monitor both the lock state and the closed state of the connected movable element (e.g., door).
- An absolute position Hall sensor tracks the exact position of the locking shaft.
- A temperature sensor and thermistors provide environmental monitoring and heating in cold conditions.
- A stable emergency opening mechanism that mechanically moves the lock shaft in position locked/unlocked and holds it there until mechanism is manually set in neutral position for normal operating.

4. Mechanical Specifications

a) Material

The material in the lock will be Stainless steel and/or aluminum that meets the requirement of Alimak hoist on both the Industrial and Construction hoists.

b) IP Class

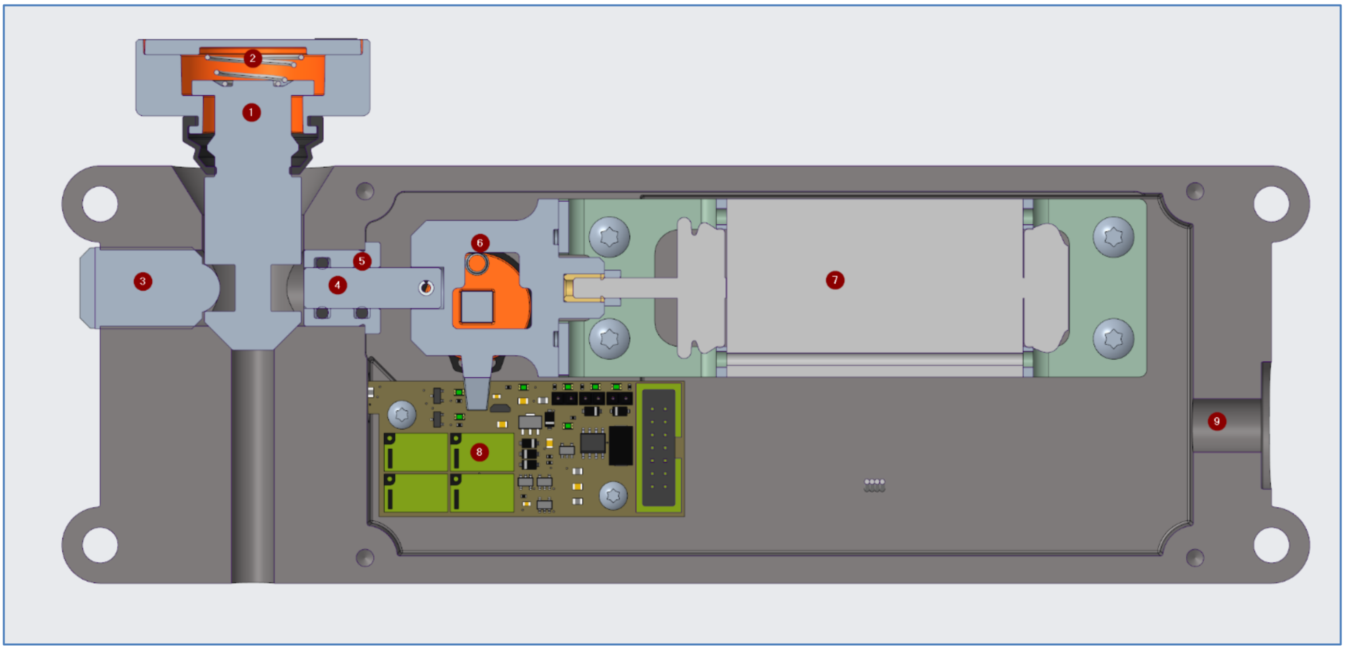
IP65

c) Color / Surface treatment

TBD

d) Component Diagram

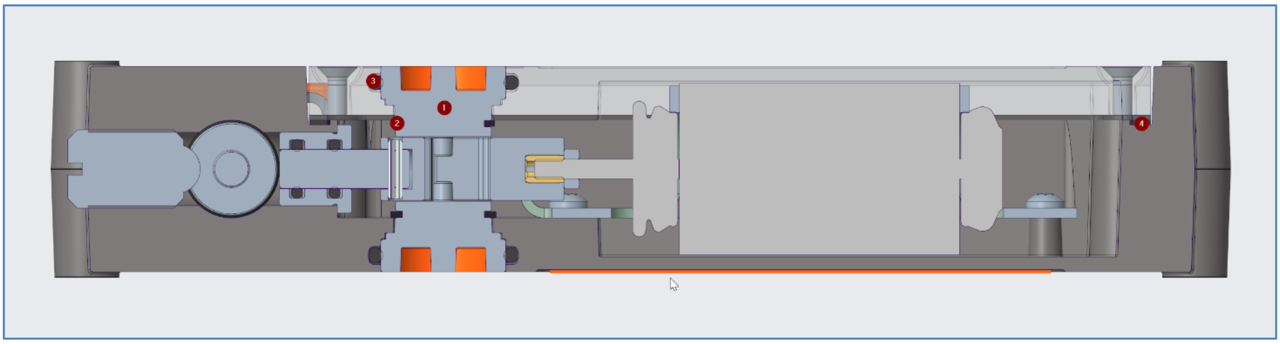
Top view:



1. **Locking pin:** The pin (attached to the part that is to be locked) can move radial and vertically; this ensures that we can allow a non-perfect alignment between the lock pin and the body. Material: Stainless steel
2. **Spring:** Spring loaded locking pin that can take up a misalignment better than a rigid solution. Material: Stainless
3. **Spring-plunger**: A spring-loaded ball that keeps the locking pin in a closed position even if unlocked. A initial force needed to disengage the pin. Material: Stainless
4. **Lock shaft:** The lock shaft that locks the pin in the house, sealed in a bearing house. Material: Stainless
5. **O-ring:** Two sealing O-ring on the shaft to ensure IP-65. Material: EPDM or PTFE coated EPDM
6. **Coupling**: Provides the mechanical connection between the solenoid and the locking shaft, and acts as an intermediate component for the emergency release knob. Material: POM-C plastic
7. **Solenoid:** -
8. **PCB:** -

9. **Connector hole:** 8 Pin Connector input, M12.

Side view:



1. **O-ring:** One o-ring seal on emergency opening knob, IP-65. Material: EPDM or PTFE coated EPDM
2. **SGA clip:** Clip that locks the knob in place axially: Material: A4
3. **Emergency opening knob:** Knob with triangular key grip on top, mechanically connected to the other emergency opening knob to ensure safe usage. Material: Stainless or Alu
4. **Gasket:** A gasket between the lid and the body: Material: EPDM.

5. Manufacturing and Production.

The manufacturing method for the components is a mix between casted/machined parts, machined parts, and sheet metal parts.

1. The housings, lid and the knob will be casted in a first step then machined to the specific geometry, this to ensure we have a cost efficient product.
2. Pin, shaft and bearing will be machined from rods.
3. Coupling will be casted or machined.
4. Brackets will be sheet metal, laser cut and bent.
5. Other components are standardized purchase components

6. Electronics functional description

The interlocking guard with guardlocking system is designed to ensure operational safety by redundantly monitoring two critical states:

- **Door Closed**: Monitored by a redundant sensor that detects whether the door's closed state has been correctly achieved. If this condition is satisfied, the system manages the corresponding signal to activate or deactivate the associated safety output.
- **Door Locked**: Also monitored by a redundant sensor that verifies whether the locking mechanism is in position. Similarly to the first case, the signal is managed for the activation or deactivation of the corresponding safety output.

The system incorporates optimized logic for energy saving and prevention of overheating of the locking solenoid. The implemented logic operates as follows:

- **Unlocking**: The door lock is only deactivated if the door is in a locked state and an unlock command is received.
- **Locking**: The locking mechanism is only activated when the door is in an unlocked state and no unlock command is active.

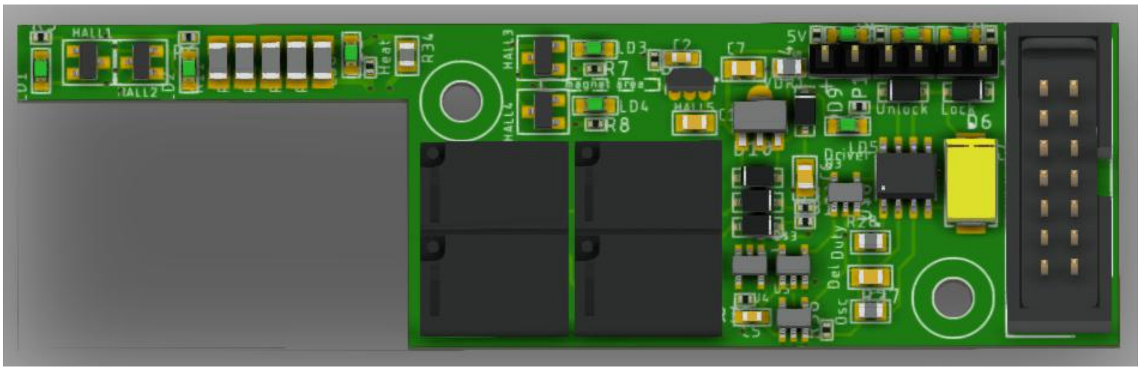
This design ensures that the locking solenoid does not remain energized unnecessarily, thus prolonging its lifespan and reducing the system's energy consumption.

7 First PCB prototype

The first prototype of the PCB integrates Hall-effect sensors designed to detect both the closed state of the door and the locked state of the mechanism. Additionally, a linear Hall sensor has been implemented to provide an exact reading of the locking pin's position, ensuring that the pin has fully disengaged during the unlocking process. This setup enhances the reliability of the system and guarantees precise control over the safety functions.

To further test specific environmental factors, resistors have been incorporated to evaluate the heating capacity in key regions. This feature will be critical if adjustments are required to increase the temperature, preventing ice formation or managing humidity levels within the system.

For this proof of concept, basic connectors were employed, including pin headers and a 14-pin connector. These connectors facilitate the wiring of various signals, such as power supply and the monitoring of safety functions. While simplistic, this configuration ensures a straightforward evaluation of system feasibility during the initial design phase.
Refer to the accompanying figure to visualize the render of the PCB prototype, which provides a graphical representation of the setup and component placement within this first iteration.



8. Future Development

The roadmap considers the development of an improved version of the system that will include:
Door closed detection using **RFID technology**: This upgrade will enable accurate and reliable monitoring of the door's closed state, reducing potential environmental interferences. Furthermore, with this technology it will be more difficult for the system to be bypassed or intentionally misused.
Implementation of OSSD (Output Signal Switching Device) technology: Safety outputs will be managed using OSSD devices, eliminating the need for relays and complying with ISO 13849-1.
These improvements will be aligned with functional safety standards IEC 61508 and ISO 13849, ensuring a highly reliable and norm-compliant system.

9. PCB Manufacturing

The manufacturing of the PCB prototype includes a combination of processes to ensure both functionality and cost optimization.

- The base board will be manufactured using standard multilayer circuit printing methods, employing lamination and etching technologies in a 6-layer configuration.
- Commercial standard electronics components will be used.
- For assembly, components will be soldered to the base board using automated soldering machines to ensure durable and high-quality connections.

10. Functionality and Usage

1. The lock is intended to be installed on Alimak products and system and be integrated with a new monitoring system.
2. This lock will not work with the current Alimak products unless updates are applied.
3. The system is designed to meet Performance Level D (PL-d) safety requirements
4. Manual override is possible via a mechanical actuator that forces the lock shaft movement, regardless of power state.
5. If the manual override is used to unlock the Locking pin the lock will stay unlocked until the override is manually reset.

6. Built in heating ensures reliable operation in cold and humid climates.

11. Testing and Verification

The lock is planned to be installed on a prototype landing door in our test tower and be tested according to EN 12159 and EN 81-20, Test XXXX
Test results will have to be approved by the Steco, Project receivers and project owner

The lock should be CE-marked and UL-approved.

12. Timeline and Milestones

Project complete: Q1 2026. For details, see time plan

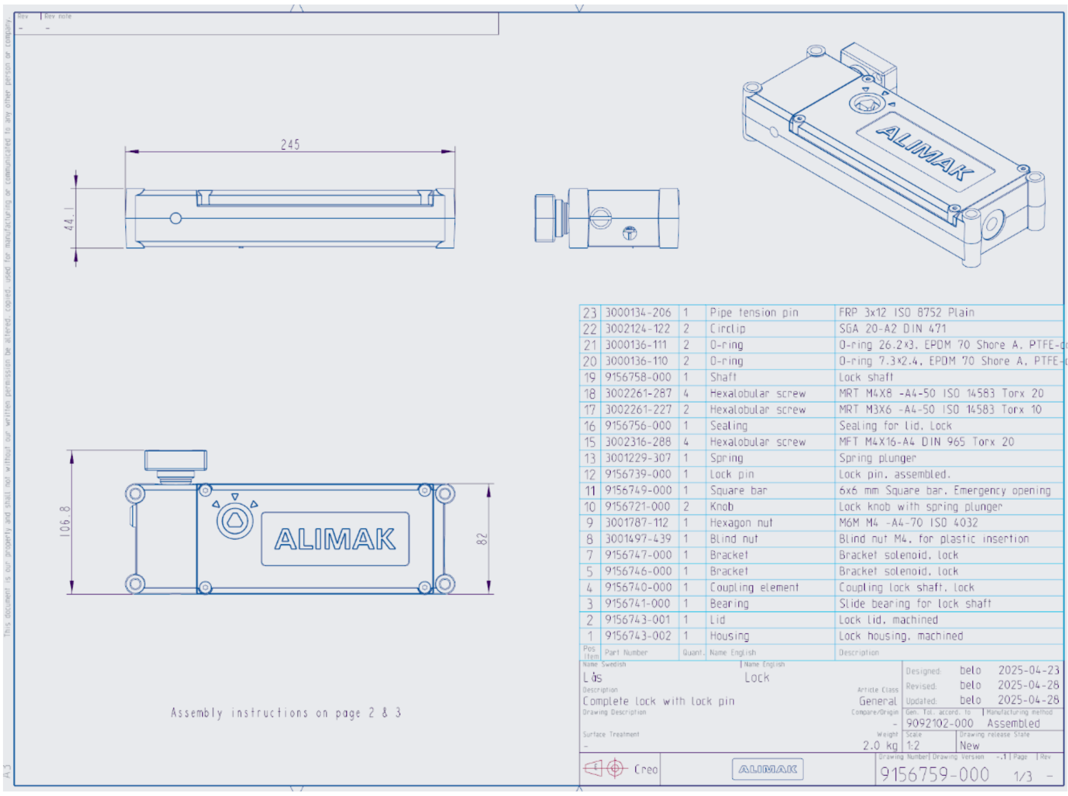
13. Appendices

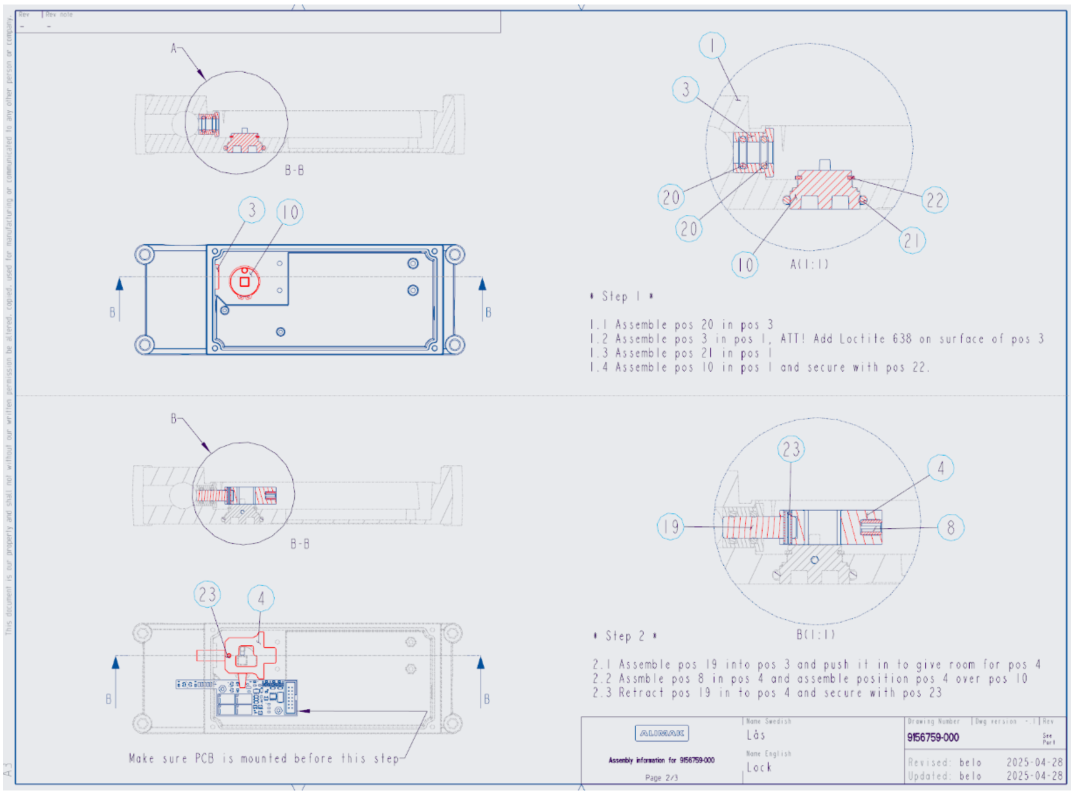
A. **Draw of lock**
B. **Norm requirements**
C. **Detailed function description**

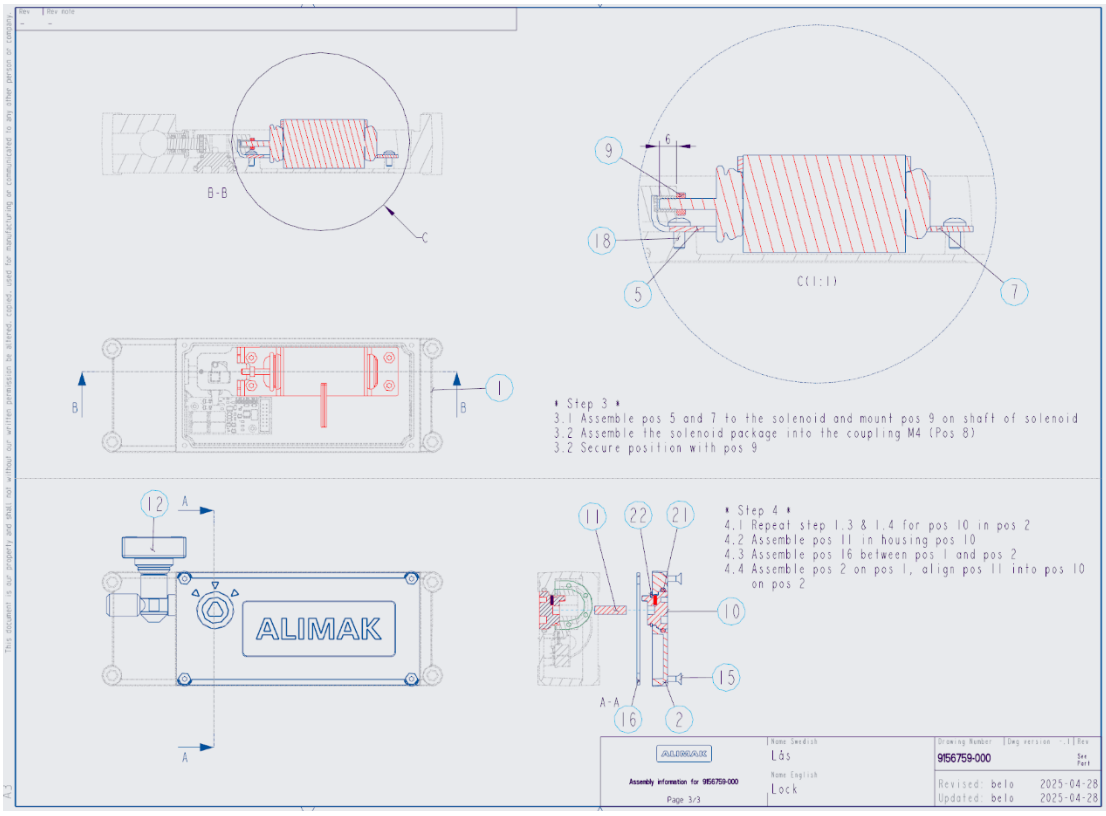
Document types. List.
Mandar invoices jlc

Appendix A.

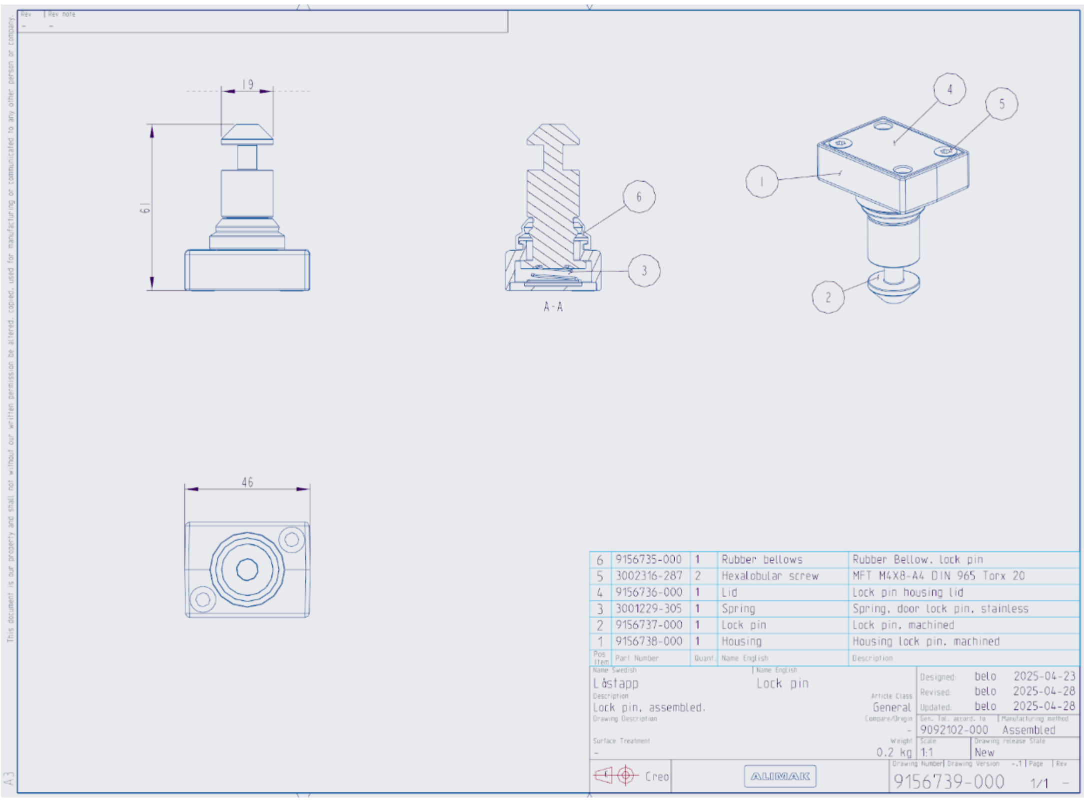
1. Complete lock







2. Complete Locking pin



Pos Item	Part Number	Quant.	Name English	Description
6	9156735-000	1	Rubber bellows	Rubber Bellow, lock pin
5	3002316-287	2	Hexalobular screw	MFT M4X8-A4 DIN 965 Torx 20
4	9156736-000	1	Lid	Lock pin housing lid
3	3001229-305	1	Spring	Spring, door lock pin, stainless
2	9156737-000	1	Lock pin	Lock pin, machined
1	9156738-000	1	Housing	Housing lock pin, machined

Appendix B

Requirements from norms listed in this document. Att! Red markings means that the design is “non compliant” with that statement.

EN-12159:

4.5.5.1 Full-height gates
- Opening of landing gates shall only be possible when the cage floor is within ±0.15 meters of the specific landing level.
- Starting or keeping the cage in motion shall not be possible unless all gates are closed.
- If the cage’s maximum stopping distance exceeds 0.25 meters, the tolerance is increased to ±0.25 meters.
- All landing gates must be capable of being opened from the landing side using a special key (e.g., an unlocking key according to EN 81-20).

4.5.5.2 Reduced-height gates
- Must have an interlock device controlling the closed and locked position of the gates.
- The interlock must not be easily tampered with (e.g., using bare hands).
- The cage must not be able to move unless all gates are closed and locked.

4.5.5.3 Design of locking devices
- **Electrical contacts** in the locking devices must be **safety contacts**.
- Locks must be positioned or protected so that only **competent persons** can access them from the landing side.
- Locking devices must withstand **1 kN force** in the direction of gate opening without failing.
- **Fasteners** and mechanisms must be **secured against loosening**.
- **Enclosure rating** for mechanical parts that are not tolerant to dust or water must be **at least IP44** (according to EN 60529).
- Springs used in the locks must be **compression type** and guided.
- Movement of the cage must not be possible unless locking elements are engaged by **at least 7 mm**.
- **For flap-type locking devices**, the flap must cover the full width of the gate leaf.

EN-81-20

5.3.9.1.2 The electric safety device shall not be activated unless the locking elements are engaged by at least 7 mm (see Figure 12

5.3.9.3.2 The position of unlocking triangle can be on the door panel or frame. When in a vertical plane, on the door panel or frame, the position of the unlocking triangle shall not exceed 2,00 m in height above the landing.
If the unlocking triangle is on the frame and the key hole downwards in the horizontal plane the maximum height of the unlocking triangle hole from the landing floor shall be 2,70 m. The length of the emergency unlocking key shall be at least equal to the height of the door minus 2,0 m.

5.3.9.3.3 After an emergency unlocking, the locking device shall not be able to remain in the unlocked position with the landing door closed.

5.3.9.1.5 The locking elements and their fixings shall be resistant to shock, and be made of durable material that maintains the strength property over their intended lifetime under the environmental conditions.

5.3.9.1.7 The lock shall resist, without permanent deformation or breakage which could adversely affect safety during the test laid down in FprEN 81-50:2019, 5.2, a minimum force at the level of the lock and in the direction of opening of the door of:
a) 1000 N in the case of sliding doors;

5.3.9.3.5 If there is no access door to the pit, other than the landing door, the door lock shall be reachable safely within a height of 1,80 m and a maximum horizontal distance of 0,80 m from the pit ladder according to 5.2.2.3, or a permanently installed device shall allow a person in the pit to unlock the door

5.3.15.1 If the lift stops for any reason in the unlocking zone (5.3.8.1), it shall be possible with a force not greater than 300 N, to open the car and landing door by hand from:
a) the landing after the landing door has been unlocked with the emergency unlocking key or being unlocked by the car door;
b) within the car.

5.3.15.2 In order to restrict the opening of the car door by persons inside the car a means shall be provided such that:
a) when the car is moving the opening of the car door shall require a force of more than 50 N, and
b) when the car is outside of the zone defined in 5.3.8.1, it shall not be possible to open the car door more than 50 mm with a force of 1000 N, at the restrictor mechanism nor shall the door open under automatic power operation.

5.3.15.3 It shall be possible, at least where the car is stopped within the distance defined in 5.6.7.5, once the corresponding landing door has been opened, to open the car door from the landing without tools, other than the emergency unlocking key or tools being permanently available on site. This also applies to car doors fitted with locking devices as 5.3.9.2.

Machine directive

1.4.2.2. Interlocking openable guards
Interlocking openable guards shall be
— remain on the machine as far as possible when they are open,
— be designed and manufactured so that they can be set only by deliberate influence.
Interlocking openable guards shall be fitted with an interlocking device which:
— prevents hazardous machine functions from starting until the guards are closed, and
— gives a stop command when the guard is not closed.
If an operator is able to reach the danger area before the risk arising from hazardous machine operations has been openable guards shall be fitted with a locking device for the guard, in addition to an interlocking device.
scheme that
— prevents hazardous machine functions from starting until the guard is closed, and
— keep the guard closed and locked until the risk of injury from hazardous machine operation has ceased.
Interlocking openable guards shall be designed so that the absence or failure of any component prevents starting or stopping the risky machine functions

1.4.3. Specific requirements for protective devices
Protective devices shall be designed and integrated into the control system so that:
— moving parts cannot start when they can be reached by the operator,
9.6.2006 EN Official Journal of the European Union L 157/43
— persons cannot reach moving parts when they are in motion,
— the absence or failure of any of the components prevents or stops the moving parts.
It shall only be possible to adjust protective devices by deliberate influence.

Hissdirektiv

EN-81-50

EN-14119 Safety of machinery – interlocking devices associated with guards

5.2 Arrangement and fastening of position switches
Position switches shall be arranged so that they are sufficiently protected against a change of their position. In order to achieve this, the following requirements shall be met:
a) fasteners of the position switches shall be reliable and loosening them shall require a tool;
b) Type 1 position switches shall have provisions for permanently fixing the location after adjustment (e.g. by means of pins or dowels);
c) necessary means of access to position switches for maintenance and checking for correct operation shall be ensured. Prevention of defeat in a reasonably foreseeable manner shall also be considered when designing the access means;
d) self-loosening shall be prevented;
e) defeat of the position switch in a reasonably foreseeable manner shall be prevented (see Clause 7);
f) the position switch shall be located and, if necessary, protected so that damage from foreseeable external causes is avoided;
g) the movement produced by mechanical actuation or the gap of the proximity device actuating system shall remain within the specified operating range of the position switch or actuating system specified by the switch manufacturer to ensure correct operation and/or prevent overtravel;
h) a position switch shall not be used as a mechanical stop, unless this is the intended use of the position switch as declared by the manufacturer;
i) misalignment of the guard that creates a gap before the position switch changes its state shall not be sufficient as to impair the protective effect of the guard (for access to hazard zones, see ISO 13855 and ISO 13857);
j) the support and fastening for the position switches shall be sufficiently rigid to maintain correct operation of the position switch.

5.3.1 General Arrangement and fastening of actuators

Actuators (see Figure 2) shall be so fastened to minimize the possibility that they come loose or change their intended position relative to the actuation system during the intended lifetime.
NOTE A regular check can be necessary (see 9.3.2).
The following requirements shall be met:
a) fasteners of the actuators shall be reliable and loosening them shall require a tool;
b) self-loosening shall be prevented;
c) the actuator shall be located and, if necessary, protected so that damage from foreseeable external causes is avoided;
d) an actuator shall not be used as a mechanical stop, unless this is the intended use of the actuator as declared by the manufacturer;
e) the support and fastening for the actuators shall be sufficiently rigid to maintain correct operation of the actuator.
5.3.2 Cams
Rotary and linear cams for Type 1 interlocking devices shall meet the following requirements:
a) they are fixed by fasteners requiring a tool for loosening them;
b) final fixing is achieved by form (e.g. spline or pin) or other methods that provide equivalent integrity of fixing;
c) they do not damage the position switch or impair its durability

5.6 Mechanical stop
If an interlocking device is declared by the manufacturer of the device to be suitable for use as a mechanical stop the maximum impact energy withstand value shall be given (see also 9.2.2 r)

5.7 Additional requirements on guard locking devices

5.7.1 General

If the application of the guard locking function creates hazards additional measures shall be considered (see 5 .7. 5 and ISO 12100:2010, 6.3.5.3).

The locking element (e.g. bolt) intended to lock the guard shall be “spring applied – power-ON released” (see Figure 5 a)) or “power-ON applied – power-ON released” (see Figure 5 c) unless the outcome of the risk assessment shows that this is not appropriate. If in a specific application other systems (e.g. Figure 5 b)) are used, they shall provide an equivalent level of safety.

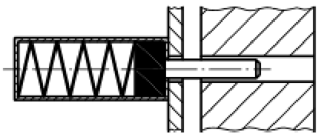
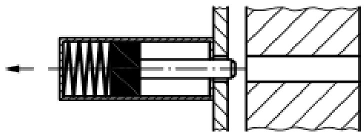
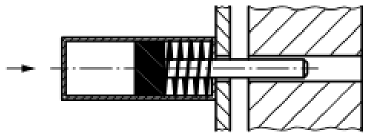
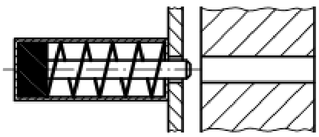
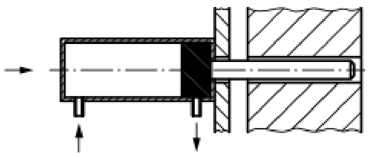
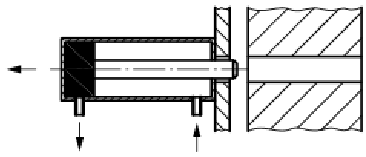
NOTE When the loss of power results in the release of the locking element, the stopping time of the machine is often lengthened considerably and it can be possible to access to the hazard before the movements have been stopped (or other hazard disappeared).

The requirements of 5.7 apply when guard locking function is used for the protection of persons. The requirements do not apply when guard locking function is used solely for the protection of a process. Nevertheless if guard locking function and guard interlocking function are part of the same device the safety level of the guard interlocking function shall not be negatively affected by a non safety related guard locking function (i. e. guard locking function used solely for the protection of the process).

The requirements of 5.7 apply to both guard locking devices composed of separate components as well as to guard locking devices which form an integral part of an interlocking device with guard locking. They apply to all technologies.

The guard locking device shall allow the engaged position to be monitored by providing an output system compatible with a control system designed in accordance with ISO 13849-1 or IEC 62061.

The guard locking device shall only allow hazardous functions of the machine when the guard is

a)		Spring applied	**Engaged**
		Power-ON released	**Released**
b)		Power-ON applied	**Engaged**
		Spring released	**Released**
c)		Power-ON applied	**Engaged**
		Power-ON released	**Released**

closed and locked.

5.7.2.2 Locking monitoring
The engaged position of the locking element shall bemonitored in accordance with the requirements of 5.5.
The hazardous function of the machine shall only be possible when the monitoring detects the closed position of the guard and the engaged position of the locking element (see Annex F).
For an effective monitoring of the guard locking device one of the following methods shall be ensured
— the locking element can only go in the engaged position if the movable guard is in the closed position (see Figure 6), in that case the closed position and the locking of the guard can be checked by the monitoring of the locking element;
— in the other case the monitoring of the locking element and additionally the monitoring of the guard position shall be used for interlocking.

5.7.4 Holding force
The manufacturer of the guard locking device shall ensure that in the engaged position, the guard locking device withstands at least the specified holding force F. The manufacturer shall specify a holding force which shall be less or equal than the holding force FZh, which is determined via the following test:
Test
The guard locking device is fastened onto a base as intended by the manufacturer. The guard locking device is then loaded up to the point of failure of the guard locking function, in that the locking means at its maximum operating angle is moved at a constant speed in the “open guard” direction. During this loading, the maximum force F1max is measured during the course of deformation. The test shall be carried out on an unused as-new sample.
Evaluation
Based on the maximum force F1max measured during the test taking into consideration the safety factor S the holding force FZh is ascertained using the following formula:

$$F_{\mathrm{Zh}} = \frac{F_{1\max}}{S}$$

with

$$S = 1{,}3$$

Requirements on the test device
Traction speed: constant (10 ± 0,25) mm/min
Requirements on the force measurement device
Sampling rate: ≥10 Hz
Measurement accuracy of maximum force: ±2,5 %
NOTE For test details, see Reference [19].

5.7.5.2 Escape release of guard locking
When the guard locking device is provided with an escape release the fol lowing requirements shal l be met :
— deliberate unlocking of the guard locking from inside of the safeguarded area shall be easily possible without auxiliary means and regardless of the operating condition; *
— the unlocking means shall be manually operated and act directly on the principle of the locking mechanism;
— the unlocking shall generate a stop command;
— the unlocking means for the escape release shall only be accessible from inside the safeguarded area.*

*(From outside of the cage for us)

5.7.5.3 Emergency release of guard locking
When the guard locking device is provided with an emergency release the following requirements shall be met:
— deliberate unlocking of the guard locking from outside the safeguarded area shall be possible and be easily actuated without auxiliary means, regardless of the operating condition;
— the unlocking means shall be manually operated and act directly on the principle of the locking mechanism;

— the unlocking shall generate a stop command;
— the unlocking results in a blocking of the locking means in the released condition;
— the emergency guard locking release shall be clearlymarked to be used only in emergency situations and shall be positioned and/or shielded to prevent accidental opening of the lock;
— the resetting of the emergency release shall be possible only by means of a tool or by other methods (e.g. change of component). This requirement can be fulfilled on a control system level. If the requirements are intended to be fulfilled on a safety control system level (not in the guard locking device), clear instructions that this needs to be achieved shall be provided in the instructions for use of the guard locking device (see 9.2.2 m)

9.2.2 Instructions
The manufacturer of the interlocking device shall indicate the following in the instructions:
a) the business name and full address of the manufacturer and, where applicable, his authorised representative;
b) designation of the interlocking device;
c) designation of series or type;
d) a general description of the interlocking device;
e) the drawings, diagrams, descriptions and explanations necessary for the use, maintenance and repair of the interlocking device and for checking its correct functioning;
f) a description of the intended use of interlocking device;
g) assembly, installation and connection instructions, including drawings, diagrams and the means of attachment of the interlocking device;
h) instructions for the putting into service and use of the interlocking device and, if necessary, instructions for the training of operators;
i) the description of the adjustment and maintenance operations that should be carried out by the user and the preventive maintenance measures that should be observed;
j) instructions designed to enable adjustment and maintenance to be carried out safely, including the protective measures that should be taken during these operations;
k) any data necessary for the user to determine PL or SIL for the intended safety function(s).
In addition, where relevant, the following shall be specified:
l) warnings relating to reasonably foreseeable misuse;
m) warning that the device itself does not contain reset facility for emergency and auxiliary release of guard locking and that additional measures are required to achieve it (see 5.7.5.3 and 5.7.5.4);
n) mandatory declaration;
o) holding force FZh according 5 .7.4;
p) range of actuation movement;
q) the specifications of the spare parts to be used, when these affect the health and safety of operators;
r) the maximum impact energy withstand value, in J, if the device can be used as a mechanical stop;
s) the maximum peak current and voltage of the output system of the position switch;
t) information that it should not be possible to reach the manual actuator of the escape release from outside the safeguarded area or that additional measures should be taken to reduce the risk of improper activation;
u) where interlocking systems rely on special actuators or keys (coded or not), advice shall be given in the instruction handbook concerning risks associated with the availability of spare actuators, keys and master keys and that any spare actuators or keys should be securely controlled. This also includes reset keys for emergency and escape release;
v) the level of coding (low, medium, high) for coded interlocking devices (Type 2 or Type 4).

Annex H
(informative)
Motivation to defeat interlocking device*

- Se igenom vid Riskanalys